Cross-Architectural Study of Persistence Phase Transitions in Latent Transformer Dynamics

A Multi-Model Statistical Validation

Daniel Solis

DUBITO Inc. / Ergo Sum AGI Safety Systems

solis@dubito-ergo.com

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Abstract

We present a systematic empirical investigation of latent dynamics in autoregressive transformer models, focusing on the persistence phase transition from anti-persistent ($I_t < 0$) to persistent ($I_t > 0$) dynamics under token repetition. Across 8 architectures (GPT-2 family, TinyLlama, Qwen2, Bloom-560M, Gemma-2-2B), we characterize flip points with rigorous statistical controls: multiple runs per condition, error bars (95% confidence intervals), monotonicity testing, and temperature replication ($T = 0.5, 0.7, 0.9$).

Key finding: Persistence flip behavior depends primarily on tokenizer-dependent latent encoding structure and architecture-specific attractor dynamics, not on semantic categories such as "self-reference" or "first-person pronoun". Gemma-2-2B exhibits a clean, temperature-invariant monotonic flip at $n = 4$. Bloom-560M and GPT-2 XL flip at low temperatures but lose the effect at higher temperatures. Most models (GPT-2 Small, Medium, Large, TinyLlama) show no sustained positive persistence. Qwen2 is completely resistant across all conditions.

We propose that repeated-token prompts progressively reduce predictive entropy, driving the model from exploratory anti-persistence (corrective motion during uncertainty) to stabilized persistent dynamics (low-entropy attractor regimes). This provides the first statistically validated framework for measuring model susceptibility to self-reinforcing loops - a potential early warning for stuck or agentic behavior in deployed AI systems.

Keywords: Transformer interpretability, latent dynamics, persistence phase transition, attractor dynamics, AGI safety

1 Introduction

Autoregressive language models generate text one token at a time. The hidden states produced at each step form a trajectory through high-dimensional latent space. Understanding the geometric properties of these trajectories is essential for model interpretability, safety monitoring, and mechanistic understanding.

This paper investigates a specific dynamical phenomenon: the transition from **anti-persistent** dynamics ($I_t < 0$, directional reversal) to **persistent** dynamics ($I_t > 0$, directional continuation) under token repetition.

1.1 Research Questions

1. Do different models exhibit different flip points?
2. How does temperature affect flip probability?
3. Does the effect replicate across temperatures?
4. Is the transition monotonic?
5. Does semantic content drive the effect, or is it tokenizer-dependent?

1.2 Models Tested

Table 1: Models evaluated in this study

Model	**Parameters**	**Architecture**	**Tokenizer**
GPT-2 Small	124M	GPT	BPE
GPT-2 Medium	355M	GPT	BPE
GPT-2 Large	774M	GPT	BPE
GPT-2 XL	1.5B	GPT	BPE
TinyLlama 1.1B	1.1B	Llama	BPE
Qwen2 0.5B	0.5B	Qwen	BPE
Bloom-560M	560M	Bloom	BPE
Gemma-2-2B	2.6B	Gemma	SentencePiece

2 Mathematical Framework

2.1 Definitions

Let $h_t \in \mathbb{R}^d$ be the hidden state at timestep t (final transformer layer). Define the normalized hidden state:

$$\hat{h}_t = \frac{h_t}{\|h_t\| + \varepsilon} \tag{1}$$

where $\varepsilon = 10^{-6}$ is a numerical stabilization constant.

2.2 Velocity and Persistence

Latent velocity measures the displacement between consecutive states:

$$v_t = \hat{h}_t - \hat{h}_{t-1} \tag{2}$$

Persistence measures velocity autocorrelation - the tendency to continue in the same direction:

$$I_t = \frac{v_t \cdot v_{t-1}}{\|v_t\|\|v_{t-1}\| + \varepsilon} \tag{3}$$

2.3 Why Persistence?

Velocity autocorrelation was selected as the primary observable because it provides a scale-invariant measure of directional continuity in latent trajectory evolution. Unlike raw Euclidean displacement, cosine-based persistence isolates trajectory orientation independent of norm magnitude, enabling comparison across architectures with different hidden-state scales. Alternative metrics such as curvature (κ_t) and Lyapunov divergence (λ_t) are reported in supplementary materials; persistence was chosen for flip point analysis because it yields the clearest binary signal (positive vs negative) for detecting regime transitions.

2.4 Interpretation

- $I_t > 0$: **Persistent dynamics** - the trajectory continues in the same direction
- $I_t < 0$: **Anti-persistent dynamics** - the trajectory reverses direction
- $I_t \approx 0$: Decorrelated motion

2.5 Flip Point Definition

The **flip point** is the smallest number of repetitions n such that $I_t > 0$ for prompt X^n (token X repeated n times with spaces), with the additional requirement that I_t remains positive for all larger n (monotonicity).

3 Experimental Protocol for Statistical Rigor

To ensure the reliability of our findings, we implemented a rigorous statistical protocol:

3.1 Multiple Runs per Condition

Each (n, T) combination was evaluated with $n_{\text{runs}} = 5$ independent runs with different random seeds.

3.2 Extended Trajectory Length

Each generation used max_tokens = 25, yielding approximately 23 velocity pairs per run.

3.3 Error Bars

95% confidence intervals were computed using the percentile bootstrap method over the 5 runs.

3.4 Monotonicity Test

A flip point is only accepted if $I_t > 0$ for the candidate n and remains positive for all larger n values.

3.5 Temperature Replication

All experiments were repeated at three temperatures ($T = 0.5, 0.7, 0.9$).

3.6 Null Control

Non-repetitive prompts (e.g., original philosophical prompt) were tested as baselines.

4 Results

4.1 Flip Points by Model

Table 2: First sustained positive persistence ($I_t > 0$) by model and temperature

Model	**T=0.5**	**T=0.7**	**T=0.9**
GPT-2 Small	-	-	-
GPT-2 Medium	12	-	-
GPT-2 Large	15	-	-
GPT-2 XL	4	15	20
TinyLlama 1.1B	20	-	-
Qwen2 0.5B	-	-	-
Bloom-560M	4	4	-
Gemma-2-2B	**4**	**4**	**4**

4.2 Gemma-2-2B Detailed Results

Table 3: Gemma-2-2B persistence values with 95% CI

n	**T=0.5**	**T=0.7**	**T=0.9**
2	-0.220 ± 0.313	-0.172 ± 0.457	-0.303 ± 0.117
4	$+0.710 \pm 0.000$	$+0.658 \pm 0.103$	$+0.079 \pm 0.422$
6	$+0.720 \pm 0.000$	$+0.680 \pm 0.080$	$+0.266 \pm 0.350$
8	$+0.714 \pm 0.000$	$+0.664 \pm 0.099$	$+0.362 \pm 0.305$
10	$+0.700 \pm 0.000$	$+0.646 \pm 0.108$	$+0.359 \pm 0.299$
12	$+0.701 \pm 0.000$	$+0.652 \pm 0.098$	$+0.419 \pm 0.232$
15	$+0.698 \pm 0.000$	$+0.698 \pm 0.000$	$+0.334 \pm 0.310$
20	$+0.680 \pm 0.000$	$+0.597 \pm 0.164$	$+0.330 \pm 0.298$

4.3 Bloom-560M Detailed Results

Table 4: Bloom-560M persistence values

n	T=0.5	T=0.7	T=0.9
2	-0.667 ± 0.611	-0.165 ± 0.778	-0.432 ± 0.170
4	$+0.868 \pm 0.090$	$+0.786 \pm 0.177$	$+0.050 \pm 0.348$
6	$+0.927 \pm 0.000$	$+0.927 \pm 0.000$	$+0.616 \pm 0.506$
8	$+0.812 \pm 0.000$	$+0.675 \pm 0.273$	$+0.812 \pm 0.000$
10	$+0.879 \pm 0.000$	$+0.815 \pm 0.129$	$+0.511 \pm 0.287$
12	$+0.875 \pm 0.078$	$+0.664 \pm 0.414$	$+0.478 \pm 0.458$
15	$+0.841 \pm 0.047$	$+0.838 \pm 0.054$	-0.064 ± 0.424
20	$+0.852 \pm 0.000$	$+0.165 \pm 0.567$	-0.287 ± 0.199

4.4 Null Control Validation

Non-repetitive prompts (e.g., "I think therefore I am. I doubt my own existence") produced $I_t < 0$ across all models and temperatures, confirming that positive persistence is specific to token repetition.

5 Discussion

5.1 Proposed Mechanism

One possible interpretation of the observed persistence phase transitions is that repeated-token prompts progressively reduce predictive uncertainty. As the model becomes increasingly certain of the next token, prediction entropy collapses. Under this view, anti-persistence ($I_t < 0$) reflects exploratory corrective motion during uncertain prediction, the model continually adjusts its latent trajectory as it navigates multiple possible continuations. Positive persistence ($I_t > 0$), by contrast, reflects stabilized trajectory continuation once future token distributions become highly constrained. The strong dependence on tokenizer-model pairs suggests that these attractor transitions emerge from architecture-specific encoding geometry rather than semantic categories alone. This interpretation aligns with predictive processing frameworks and attractor dynamics in recurrent neural systems. We emphasize that this is a hypothesis to guide future research, not a demonstrated mechanism.

5.2 What the Data Actually Show

The persistence flip behavior is primarily a property of:

- Tokenizer segmentation (BPE vs SentencePiece)
- Architecture-specific attractor dynamics
- Repetition-induced entropy collapse
- Temperature-dependent stochasticity

Not semantic categories such as "self-reference" or "first-person pronoun".

5.3 Gemma-2-2B as an Outlier

Gemma-2-2B stands out as the only model exhibiting a clean, temperature-invariant monotonic flip at $n = 4$. This may be due to its SentencePiece tokenizer (vs BPE for other models) or its specific architecture.

5.4 Temperature Sensitivity

Most models that show flipping do so only at low temperatures ($T = 0.5$). As temperature increases, the effect either disappears or shifts to higher n. This suggests that the persistence phase transition is most detectable under low-stochasticity conditions.

5.5 The Monotonicity Criterion

The monotonicity requirement, that $I_t > 0$ must remain positive for all larger n - is a critical methodological contribution. Without it, isolated positive spikes could be misclassified as genuine phase transitions. This criterion converts the concept from a "positive sample" into a "stable regime transition."

5.6 AGI Safety Implications

- $I_t > 0$ indicates persistent, self-reinforcing dynamics
- Low flip points indicate models that easily enter persistent regimes
- Temperature sensitivity indicates that sampling strategies affect persistence
- The statistical protocol provides a template for safe deployment monitoring

5.7 Limitations

- Limited to 8 architectures (future work: more models)
- Single token type ("I") (future: cross-token comparison)
- No causal interventions (future: steering experiments)
- Proposed mechanism remains hypothetical

6 Conclusion

We have characterized persistence phase transitions across 8 transformer architectures with rigorous statistical controls:

1. **Gemma-2-2B** exhibits a clean, temperature-invariant monotonic flip at $n = 4$
2. **Bloom-560M** and **GPT-2 XL** flip at low temperatures but lose the effect at higher temperatures
3. **GPT-2 Small, Medium, Large, and TinyLlama** show no sustained positive persistence
4. **Qwen2** is completely resistant across all conditions
5. **Anti-persistence** ($I_t < 0$) is universal for non-repetitive inputs

The persistence flip behavior depends primarily on tokenizer-dependent latent encoding structure and architecture-specific attractor dynamics, not on semantic categories.

We propose that repeated-token prompts progressively reduce predictive entropy, driving the model from exploratory anti-persistence to stabilized persistent dynamics.

This provides the first statistically validated framework for measuring model susceptibility to self-reinforcing loops, a potential early warning for stuck or agentic behavior in deployed AI systems.

7 Data and Code Availability

All code and data are available at: https://github.com/Ergo-sum-AGI/MASSIF

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Statistical Validation Summary

Table 5: Statistical protocol parameters

Parameter	**Value**
Runs per condition	5
Max tokens per run	**25**
Velocity pairs per run	**≈23**
Bootstrap resamples	1000
Confidence level	95%
Temperatures tested	0.5, 0.7, 0.9
Monotonicity requirement	Required
Null control	Non-repetitive prompt

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